

Date _____

Belgrade Air Pollution Data

SO ₂	PM ₁₀	NO ₂	NO _x	CO	NO	PM _{2.5}	P	T	RH
7.35	36.92	45.49	121.47	0.88	53.37	32.32	1010.22	6.20	90.07
20.87	89.90	70.12	414.02	2.04	224.31	77.31	1012.67	8.19	82.54
10.93	64.43	38.83	133.12	1.30	61.54	59.86	1012.87	10.04	82.36
15.21	42.57	55.74	198.03	1.12	92.96	37.21	1012.73	11.14	75.14
13.99	68.64	63.15	292.04	1.54	149.28	60.49	1006.57	9.09	77.98
18.44	64.26	67.04	304.09	1.45	154.67	48.42	1000.63	9.81	68.07
12.41	26.51	50.49	123.45	0.79	47.76	16.71	994.21	13.61	57.51
5.11	16.32	33.08	69.96	0.66	24.03	14.59	1003.62	5.07	80.41
6.21	25.14	26.58	46.63	0.72	13.05	23.56	1010.94	3.03	61.34
6.44	39.86	32.69	81.54	0.83	31.92	36.60	1004.31	1.90	89.08

SO₂ (Sulfur Dioxide)

$$\text{mean} = \frac{7.35 + 20.87 + 10.93 + 15.21 + 13.99 + 18.44 + 12.41 + 5.11 + 6.21 + 6.44}{10}$$

$$\text{mean} = 11.696$$

$$\text{std} = \frac{(7.35 - 11.696)^2 + (20.87 - 11.696)^2 + (10.93 - 11.696)^2 + (15.21 - 11.696)^2 + (13.99 - 11.696)^2 + (18.44 - 11.696)^2 + (12.41 - 11.696)^2 + (5.11 - 11.696)^2 + (6.21 - 11.696)^2 + (6.44 - 11.696)^2}{10 - 1}$$

$$= \sqrt{\frac{268.34}{9}} = 5.460$$

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Sorted: 5.11, 6.21, 6.44, 7.35, 10.93, 12.41, 13.99, 15.21, 18.44, 20.87

$$\min = 5.11$$

$$Q_1 = 6.44$$

$$Q_2 = \frac{10.93 + 12.41}{2} = 11.67$$

$$Q_3 = 15.21$$

$$\max = 20.87$$

$$IQR = 15.21 - 6.44 = 8.77$$

Outlier fences

$$\text{Lower fence} = Q_1 - 1.5 \times IQR$$

$$= 6.44 - 1.5 \times 8.77 = -6.715$$

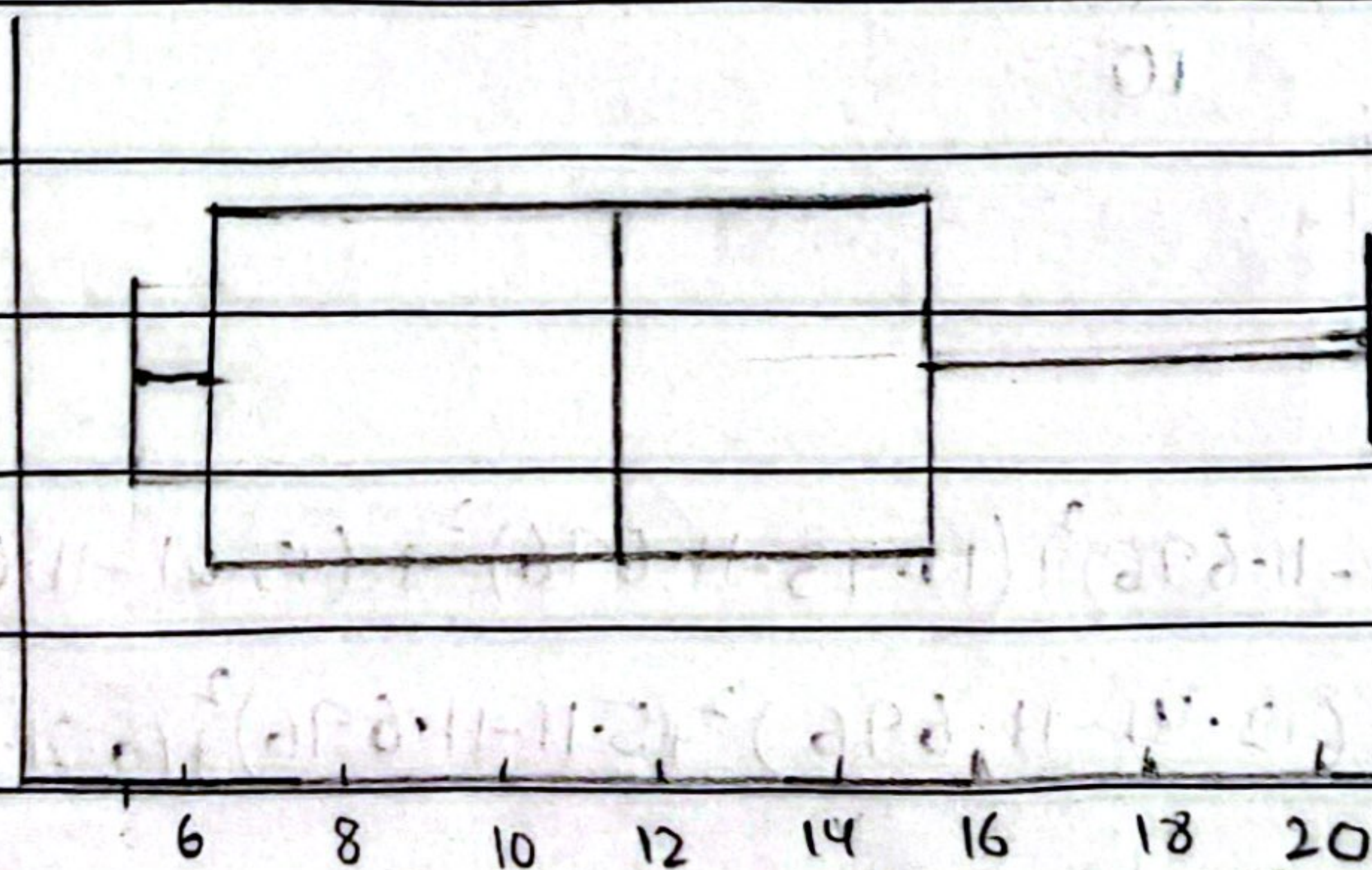
$$\text{Upper fence} = Q_3 + 1.5 \times IQR$$

$$15.21 + 1.5 \times 8.77 = 28.865$$

Any value < -6.715 or > 28.865 is an outlier

No outlier detected

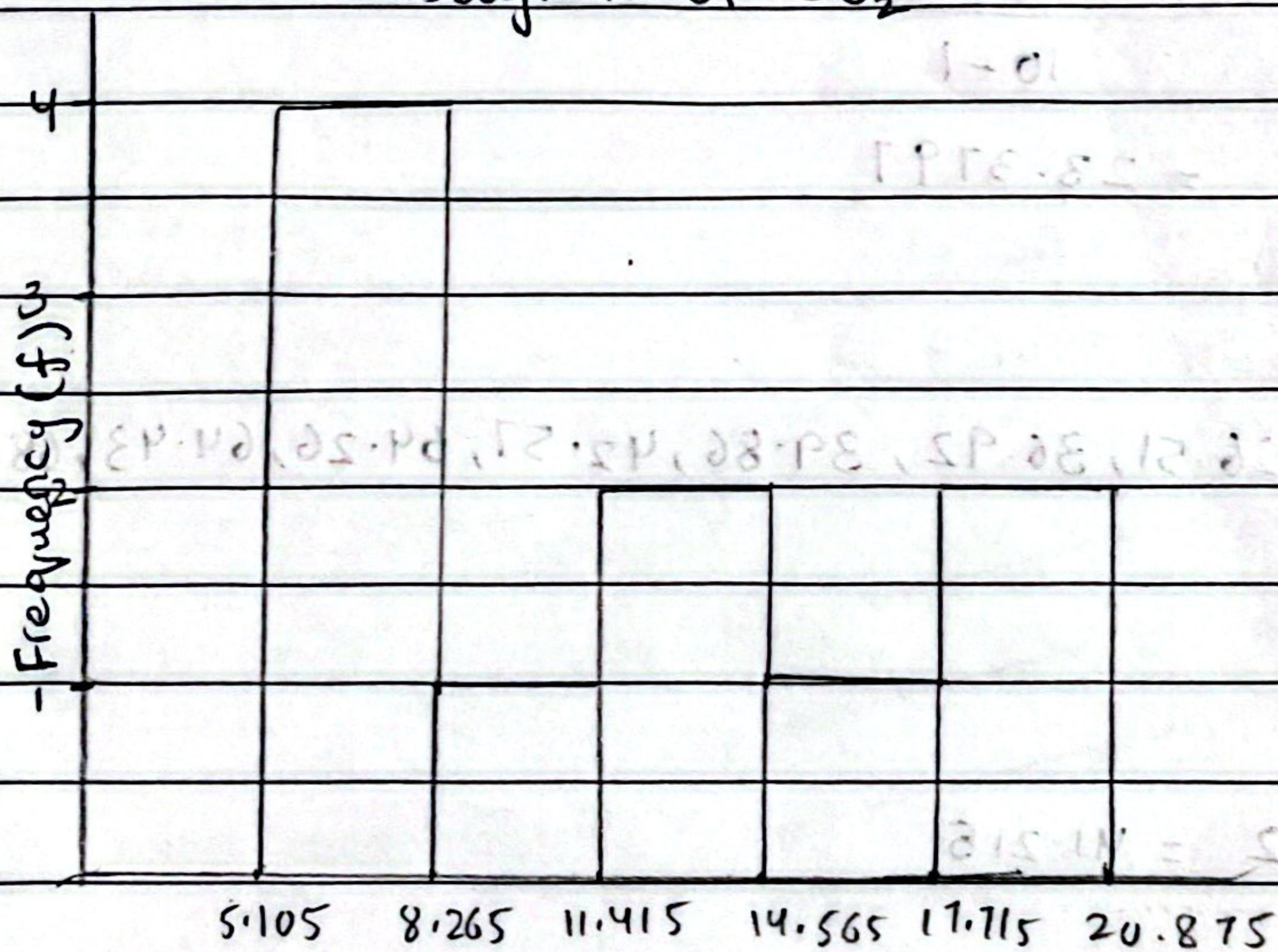
Boxplot of SO_2



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Class Limits	Class Boundaries	f
5.11 - 8.26	5.105 - 8.265	4
8.27 - 11.42	8.265 - 11.415	1
11.43 - 14.58	11.415 - 14.565	2
14.59 - 17.74	14.565 - 17.715	1
17.75 - 20.9	17.715 - 20.875	2

Histogram of SO_2



Date _____

PM₁₀ (Particulate Matter)

$$\text{mean} = \frac{36.92 + 89.90 + 64.43 + 42.57 + 68.64 + 64.26 + 26.51 + 16.32 + 25.14 + 39.86}{10}$$

$$\text{mean} = 47.455$$

$$\begin{aligned} \text{std} = & \frac{(36.92 - 47.455)^2 + (89.90 - 47.455)^2 + (64.43 - 47.455)^2 + (42.57 - 47.455)^2}{10 - 1} \\ & + (68.64 - 47.455)^2 + (64.26 - 47.455)^2 + (26.51 - 47.455)^2 + (16.32 - 47.455)^2 \\ & + (25.14 - 47.455)^2 + (39.86 - 47.455)^2 \end{aligned}$$

$$= \sqrt{\frac{4919.51}{9}} = 23.3797$$

Sorted: 16.32, 25.14, 26.51, 36.92, 39.86, 42.57, 64.26, 64.43, 68.64, 89.90

$$\text{min} = 16.32$$

$$Q1 = 26.51$$

$$Q2 = \frac{36.92 + 39.86}{2} = 41.215$$

$$Q3 = 64.43$$

$$\text{max} = 89.90$$

$$\text{IQR} = 64.43 - 26.51 = 37.92$$

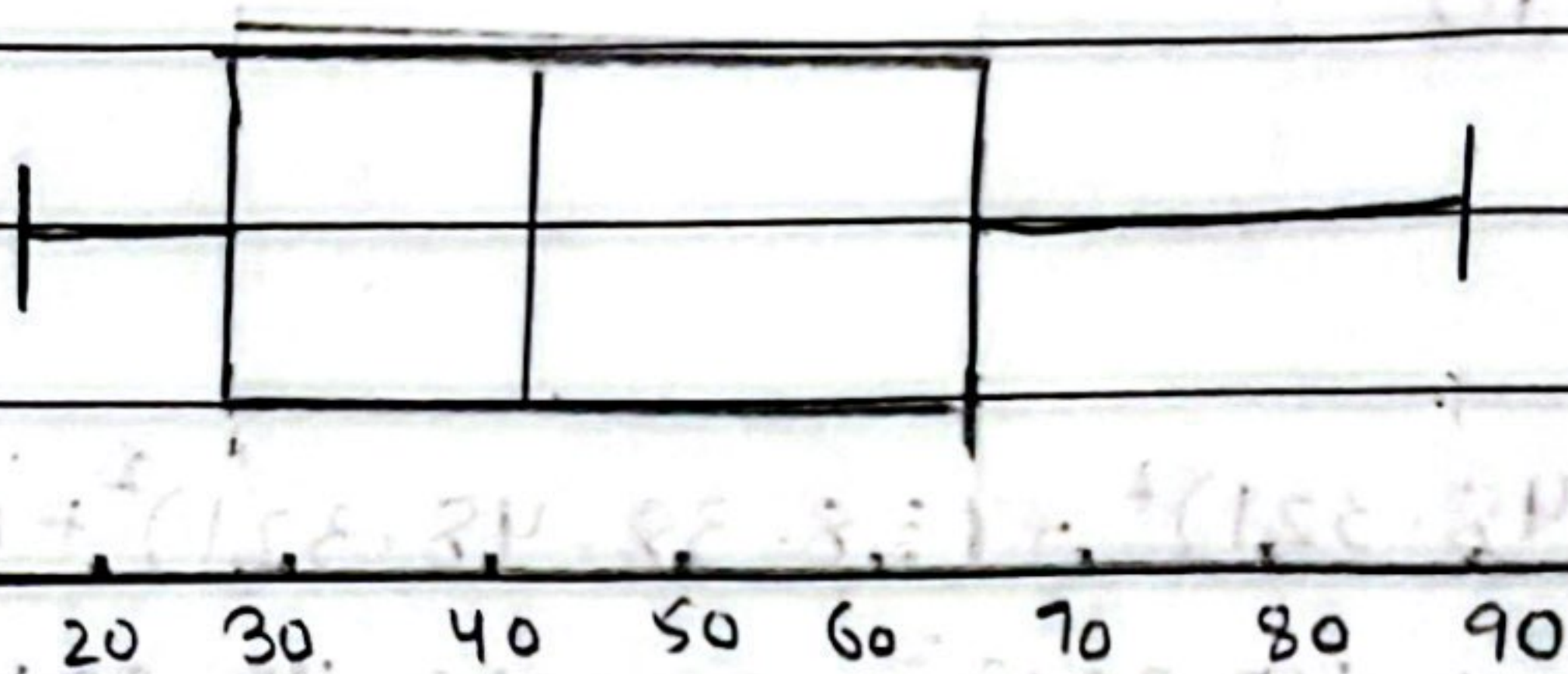
$$\begin{aligned} \text{lower fence} &= 26.51 - 1.5 \times 37.92 \\ &= -30.37 \end{aligned}$$

$$\begin{aligned} \text{upper fence} &= 64.43 + 1.5 \times 37.92 \\ &= 121.31 \end{aligned}$$

Any value < -30.37 or > 121.31 is an outlier $\rightarrow$ No outlier detected

Date _____

Boxplot of PM_{10}

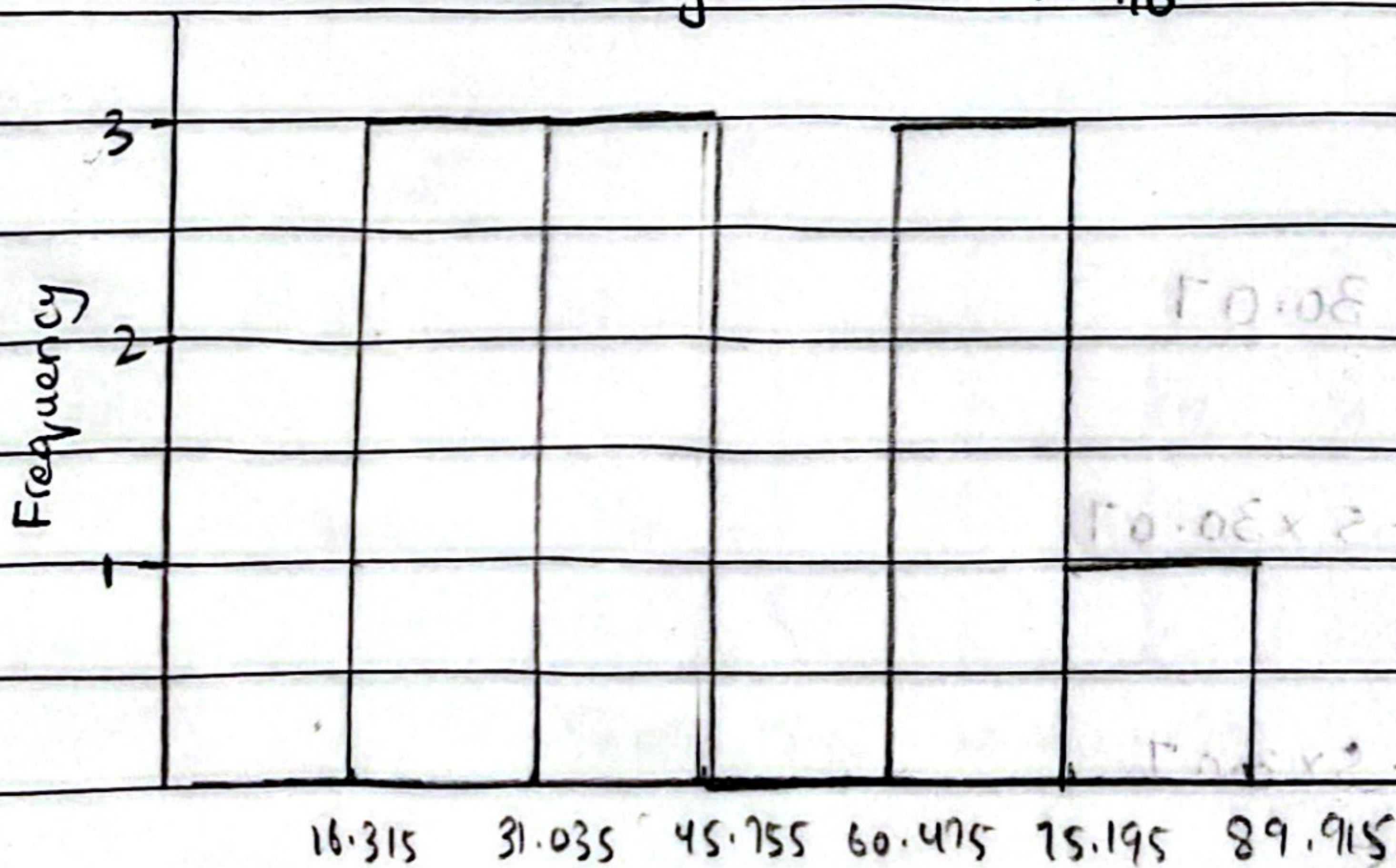


Class Limits	Class Boundaries	f
16.32 - 31.03	16.315 - 31.035	3
31.04 - 45.75	31.035 - 45.755	3
45.76 - 60.47	45.755 - 60.475	0
60.48 - 75.19	60.475 - 75.195	3
75.20 - 89.91	75.195 - 89.915	11

$$C.W = \frac{89.90 - 16.32}{5}$$

$$= 14.716$$

Histogram of PM_{10}



NO_2 (Nitrogen Dioxide)

$$\text{mean} = \frac{45.49 + 70.12 + 38.83 + 55.74 + 63.15 + 67.04 + 50.49 + 33.08 + 26.58 + 32.69}{10}$$

10

$$\text{mean} = 48.321$$

$$\begin{aligned} \text{std} = & \frac{(45.49 - 48.321)^2 + (70.12 - 48.321)^2 + (38.83 - 48.321)^2 + (55.74 - 48.321)^2}{10 - 1} \\ & + (63.15 - 48.321)^2 + (67.04 - 48.321)^2 + (50.49 - 48.321)^2 \\ & + (33.08 - 48.321)^2 + (26.58 - 48.321)^2 + (32.69 - 48.321)^2 \end{aligned}$$

10 - 1

$$= \sqrt{\frac{2152.624}{9}} = 15.465$$

$$\text{Sorted: } 26.58, 32.69, 33.08, 38.83, 45.49, 50.49, 55.74, 63.15, 67.04, 70.12$$

$$\text{min} = 26.58$$

$$Q1 = 33.08$$

$$Q2 = \frac{45.49 + 50.49}{2} = 47.99$$

$$Q3 = 63.15$$

$$\text{max} = 70.12$$

$$\text{IQR} = 63.15 - 33.08 = 30.07$$

$$\text{lower fence} = 33.08 - 1.5 \times 30.07$$

$$= -12.025$$

$$\text{upper fence} = 63.15 + 1.5 \times 30.07$$

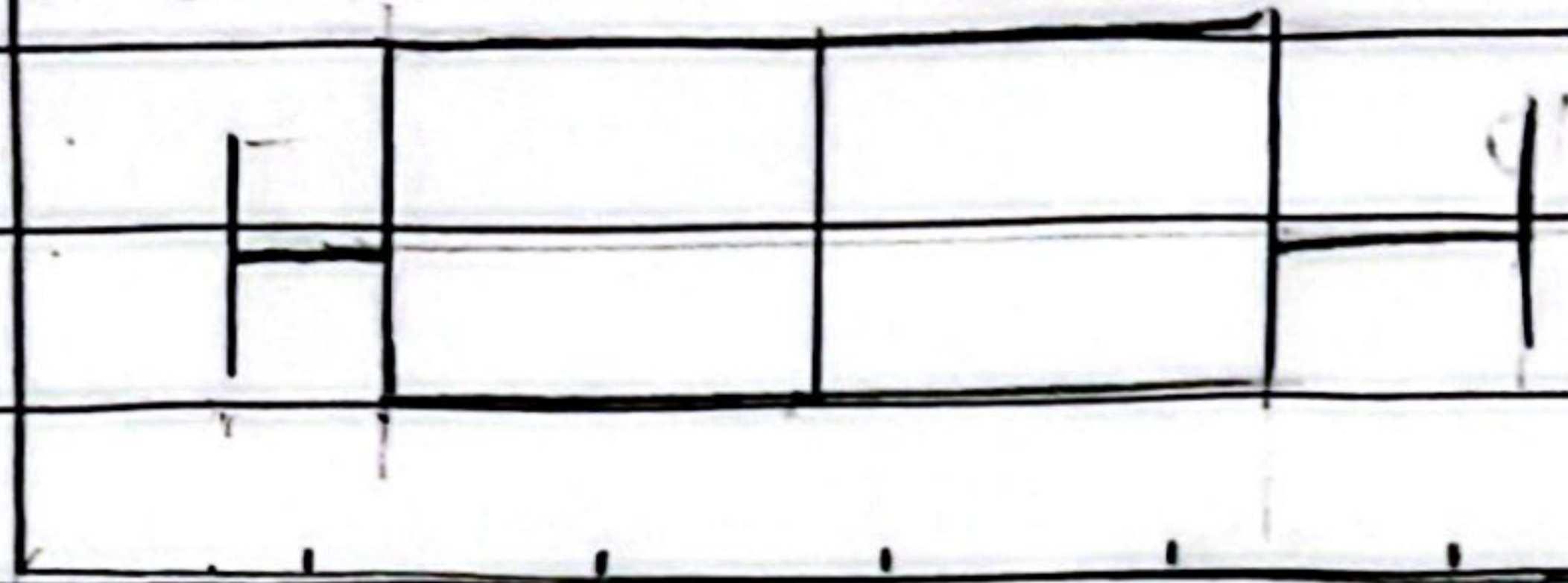
$$= 108.255$$

Any value < -12.025 or > 108.255 is an outlier

• No outlier detected.

Date _____

Boxplot of NO₂



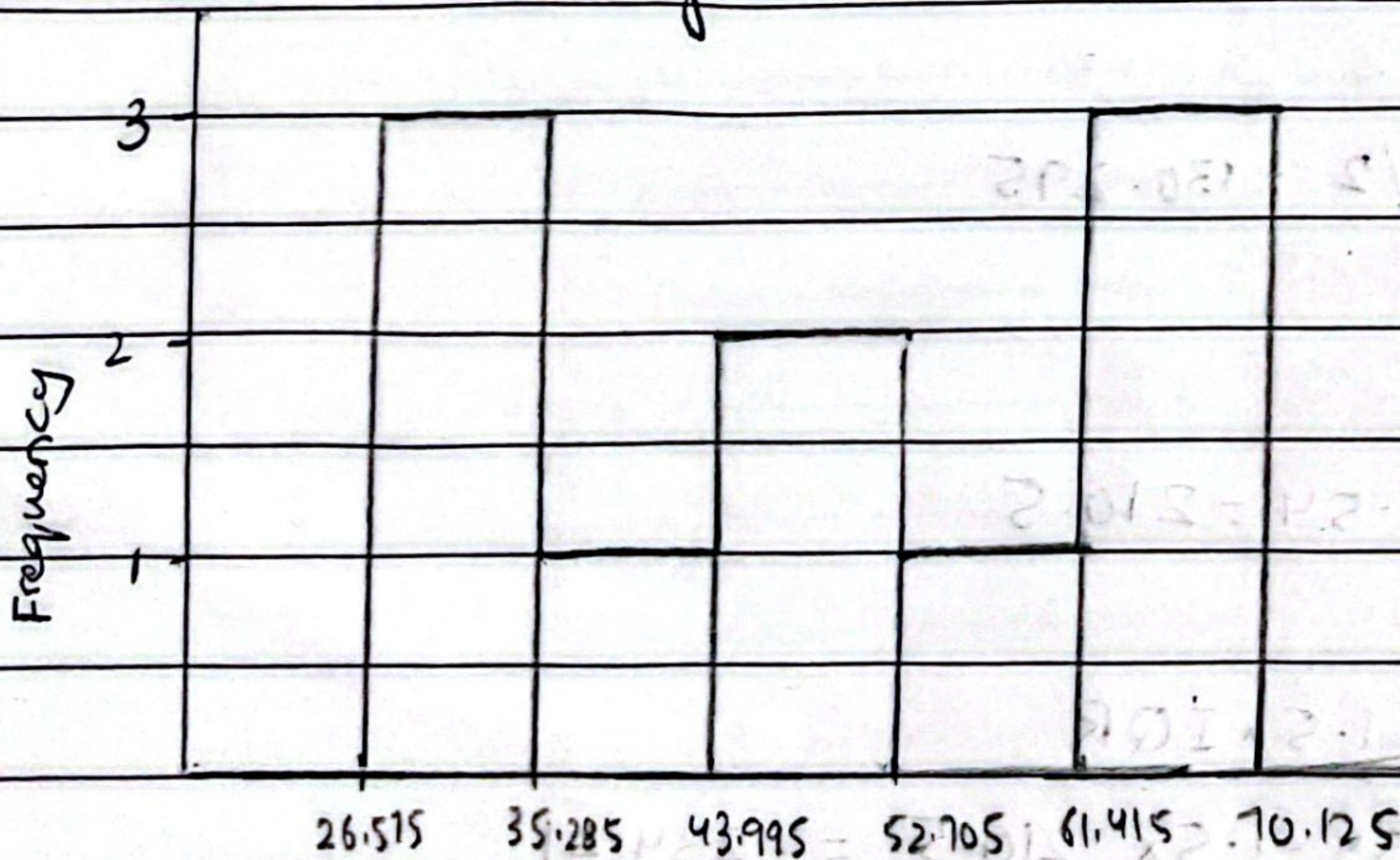
Class Limits	Class Boundaries	Frequency
26.58-35.28	26.515-35.285	3
35.29-43.99	35.285-43.995	1
44.00-52.70	43.995-52.705	2
52.71-61.41	52.705-61.415	1
61.42-70.12	61.415-70.125	3

$$C.W = 70.12 - 26.58$$

$$= 43.54$$

$$= 8.708$$

Histogram of NO₂



Date _____

NO_x (Nitrogen Oxides)

$$\text{mean} = \frac{127.47 + 414.02 + 133.12 + 198.03 + 292.04 + 304.09 + 123.45 + 69.96 + 46.63 + 81.54}{10}$$

$$\text{mean} = 179.035$$

$$\begin{aligned} \text{std} = & \frac{(127.47 - 179.035)^2 + (414.02 - 179.035)^2 + (133.12 - 179.035)^2}{10-1} \\ & + (198.03 - 179.035)^2 + (292.04 - 179.035)^2 + (304.09 - 179.035)^2 \\ & + (123.45 - 179.035)^2 + (69.96 - 179.035)^2 + (46.63 - 179.035)^2 + \\ & (81.54 - 179.035)^2 \end{aligned}$$

$$= \sqrt{\frac{130778.1867}{9}} = 120.544$$

Sorted: 46.63, 69.96, 81.54, 123.45, 127.47, 133.12, 198.03, 292.04, 304.09, 414.02

$$\text{min} = 46.63$$

$$Q1 = 81.54$$

$$Q2 = \frac{127.47 + 133.12}{2} = 130.295$$

$$Q3 = 292.04$$

$$\text{max} = 414.02$$

$$\text{IQR} = 292.04 - 81.54 = 210.5$$

Outlier Fences

$$\begin{aligned} \text{lower fence} &= Q1 - 1.5 \times \text{IQR} \\ &= 81.54 - 1.5 \times 210.5 = -234.21 \end{aligned}$$

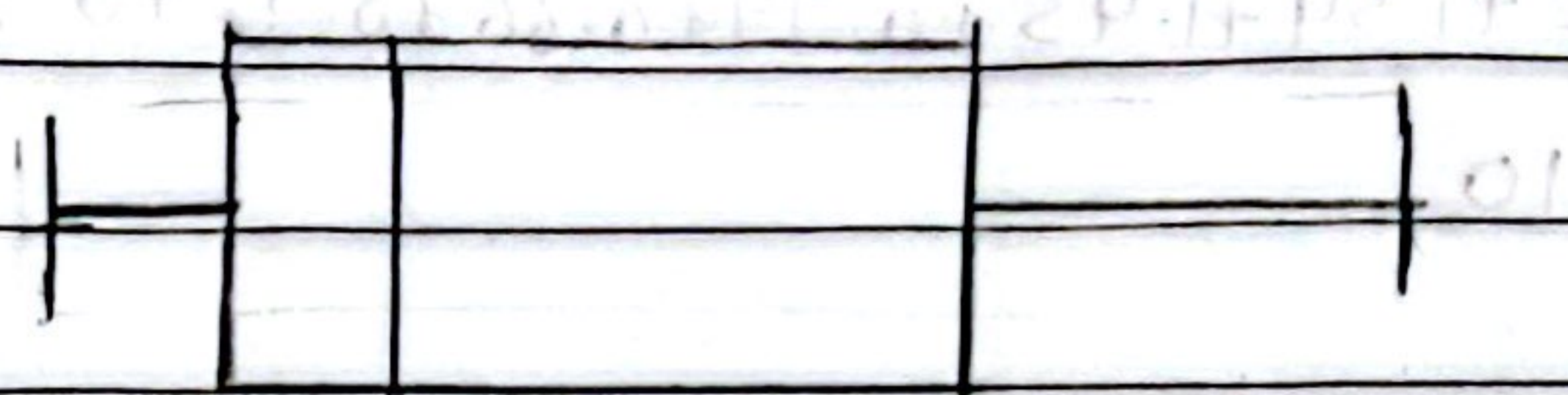
$$\begin{aligned} \text{upper fence} &= Q3 + 1.5 \times \text{IQR} \\ &= 292.04 + 1.5 \times 210.5 = 607.79 \end{aligned}$$

Any value < -234.21 or > 607.79 is an outlier

No outlier detected.

Date _____

Boxplot of NO_x



50 100 150 200 250 300 350 400 450

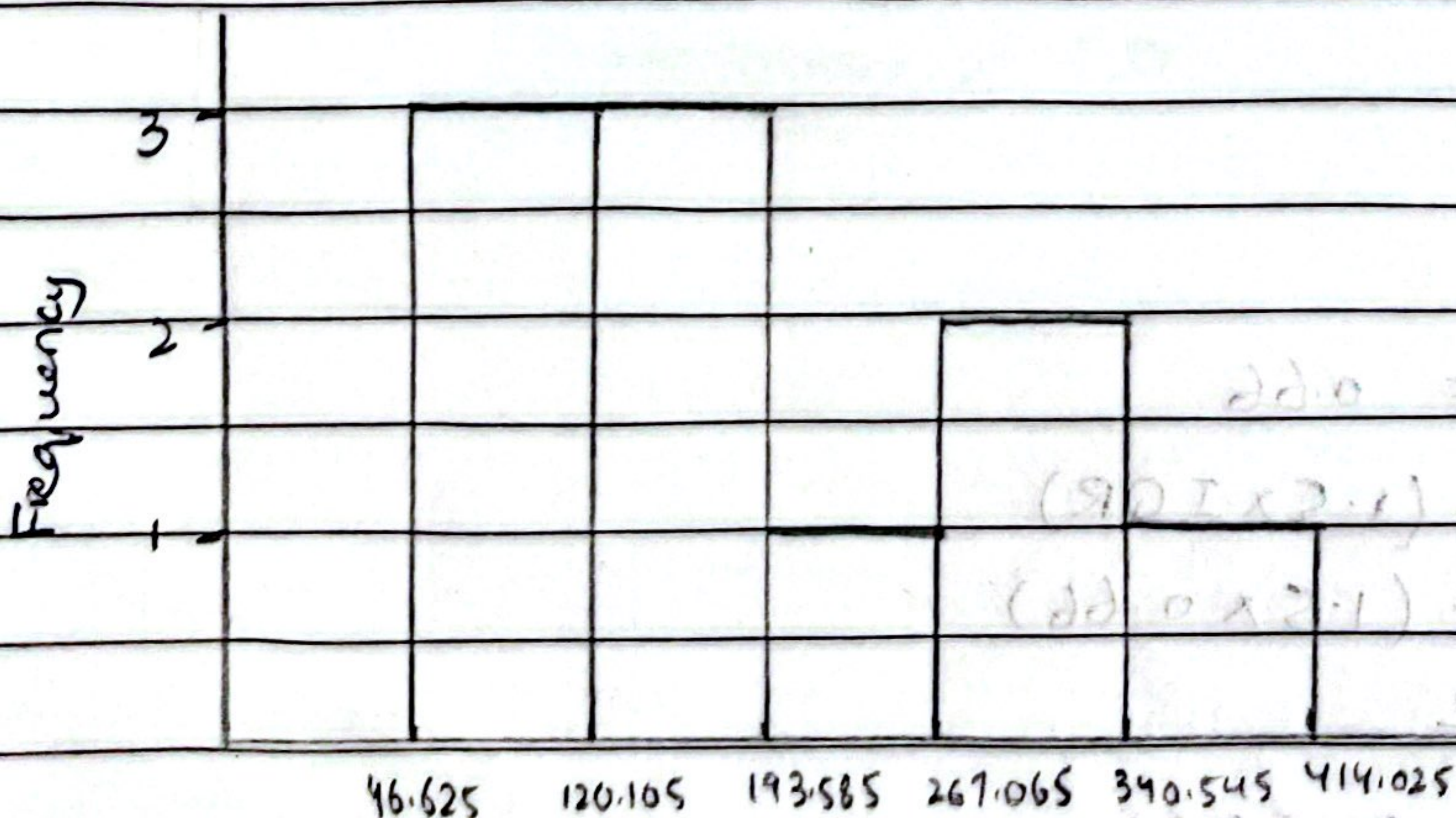
Class Limits	Class Boundaries	Frequency
46.63 - 120.10	46.625 - 120.105	3
120.11 - 193.58	120.105 - 193.585	3
193.59 - 267.06	193.585 - 267.065	1
267.07 - 340.54	267.065 - 340.545	2
340.55 - 414.02	340.545 - 414.025	1

$$C.W = 414.02 - 46.63$$

5

$$P = 73.478$$

Histogram of NO_x



46.625 120.105 193.585 267.065 340.545 414.025

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CO (Carbon Monoxide)

2014 to July 2019

$$\text{mean} = \frac{0.88 + 2.04 + 1.30 + 1.12 + 1.54 + 1.45 + 0.79 + 0.66 + 0.72 + 0.83}{10}$$

$$\text{mean} = 1.133$$

$$\begin{aligned} \text{std} = & (0.88 - 1.133)^2 + (2.04 - 1.133)^2 + (1.30 - 1.133)^2 + (1.12 - 1.133)^2 \\ & + (1.54 - 1.133)^2 + (1.45 - 1.133)^2 + (0.79 - 1.133)^2 + (0.66 - 1.133)^2 + \\ & (0.72 - 1.133)^2 + (0.83 - 1.133)^2 \end{aligned}$$

$$= \frac{1.78461}{9} = 0.445$$

Sorted: 0.66, 0.72, 0.79, 0.83, 0.88, 1.12, 1.30, 1.45, 1.54, 2.04

$$\text{min} = 0.66$$

$$Q1 = 0.79$$

$$Q2 = 1.00$$

$$Q3 = 1.45$$

$$\text{max} = 2.04$$

$$\text{IQR} = 1.45 - 0.79 = 0.66$$

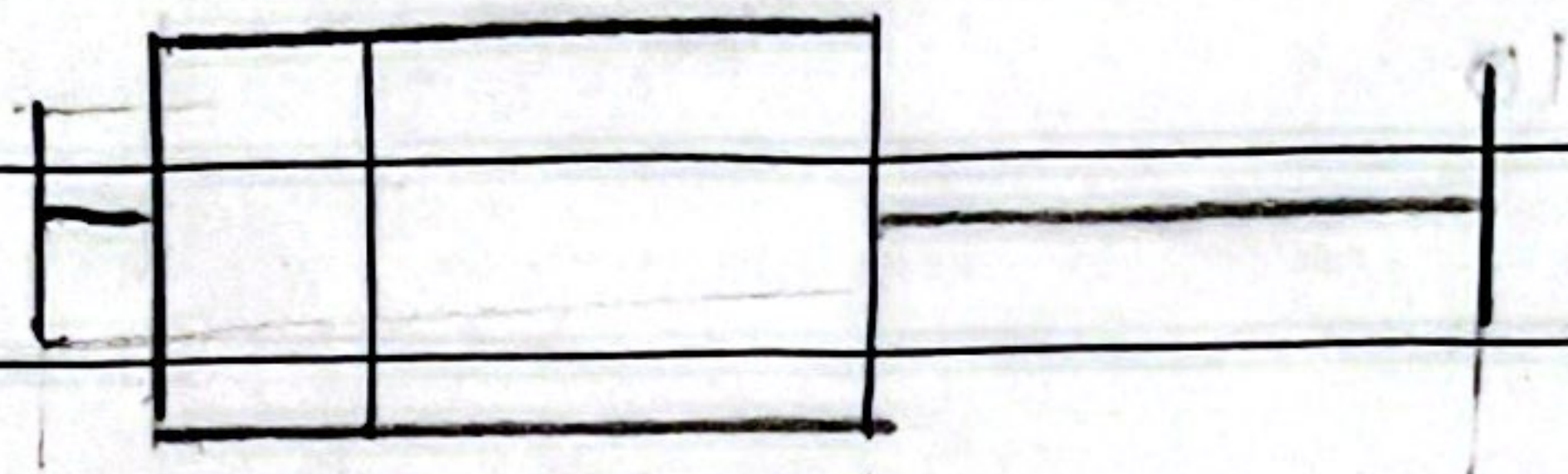
$$\begin{aligned} \text{Lower fence} &= Q1 - (1.5 \times \text{IQR}) \\ &= 0.79 - (1.5 \times 0.66) \\ &= -0.2 \end{aligned}$$

$$\begin{aligned} \text{Upper fence} &= Q3 + (1.5 \times \text{IQR}) \\ &= 1.45 + (1.5 \times 0.66) \\ &= 2.44 \end{aligned}$$

Any value < -0.2 or > 2.44 is an outlier.
No outlier detected.

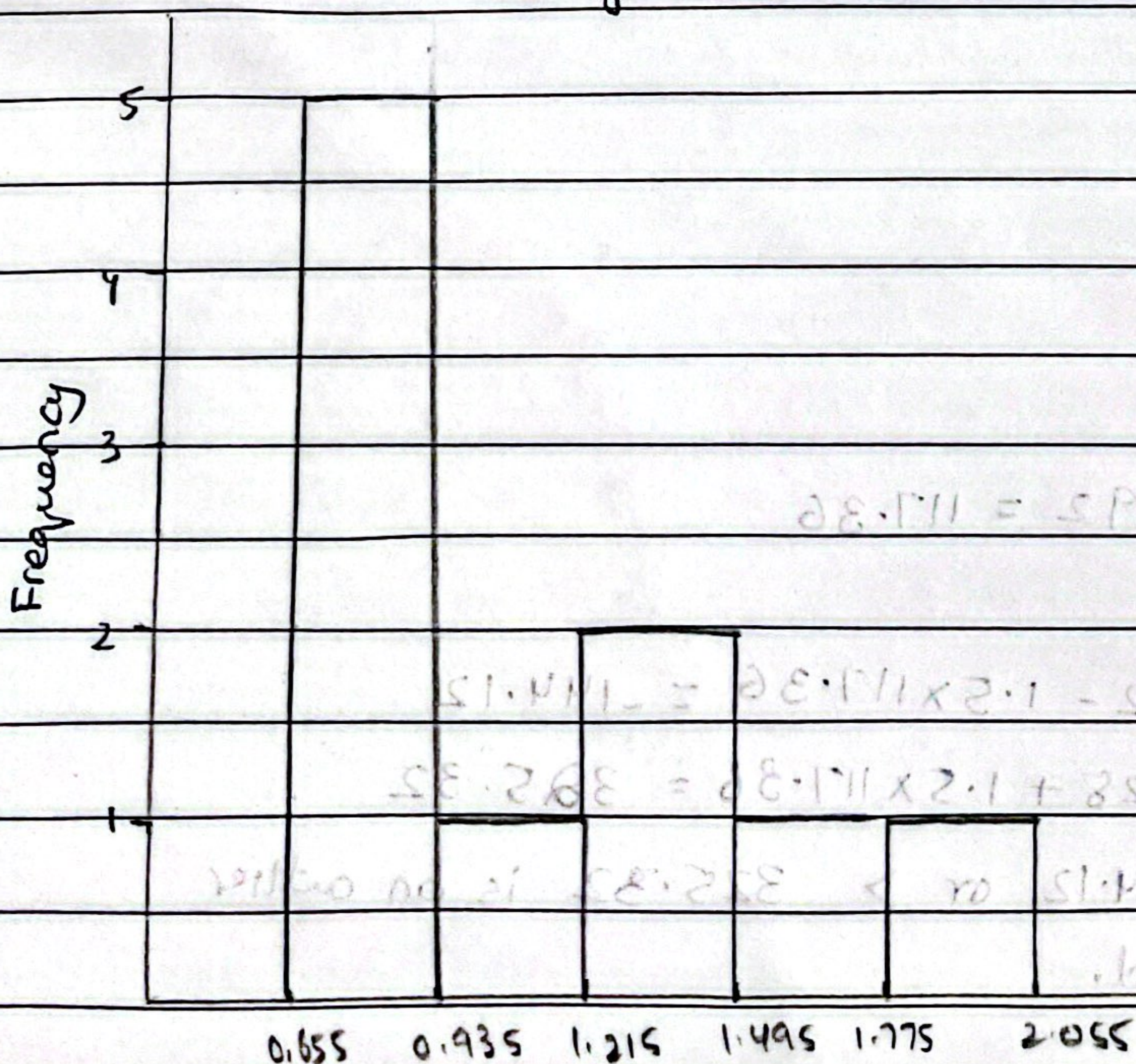
Date _____

Boxplot of CO



Class Limits	Class Boundaries	Frequency
0.66 - 0.93	0.555 - 0.935	5
0.94 - 1.21	0.935 - 1.215	11
1.22 - 1.49	1.215 - 1.495	2
1.50 - 1.77	1.495 - 1.775	1
1.78 - 2.05	1.775 - 2.055	1

Histogram of CO



Date _____

NO (Nitric Oxide)

$$\text{mean} = \frac{53.37 + 224.31 + 61.54 + 92.96 + 149.28 + 154.67 + 47.76 + 24.03 + 13.05 + 31.92}{10}$$

$$\text{mean} = 85.289$$

$$\begin{aligned} \text{std} = & \sqrt{(53.37 - 85.289)^2 + (224.31 - 85.289)^2 + (61.54 - 85.289)^2 + \\ & (92.96 - 85.289)^2 + (149.28 - 85.289)^2 + (154.67 - 85.289)^2 + \\ & (47.76 - 85.289)^2 + (24.03 - 85.289)^2 + (13.05 - 85.289)^2 + \\ & (31.92 - 85.289)^2} \end{aligned}$$

$$\text{std} = \sqrt{\frac{43104.9}{9}} = 69.206$$

Sorted: 13.05, 24.03, 31.92, 47.76, 53.37, 61.54, 92.96, 149.28, 154.67, 224.31

$$\text{min} = 13.05$$

$$Q1 = 31.92$$

$$Q2 = 57.455$$

$$Q3 = 149.28$$

$$\text{max} = 224.31$$

$$\text{IQR} = 149.28 - 31.92 = 117.36$$

Outlier Fences

$$\text{Lower fence} = 31.92 - 1.5 \times 117.36 = -144.12$$

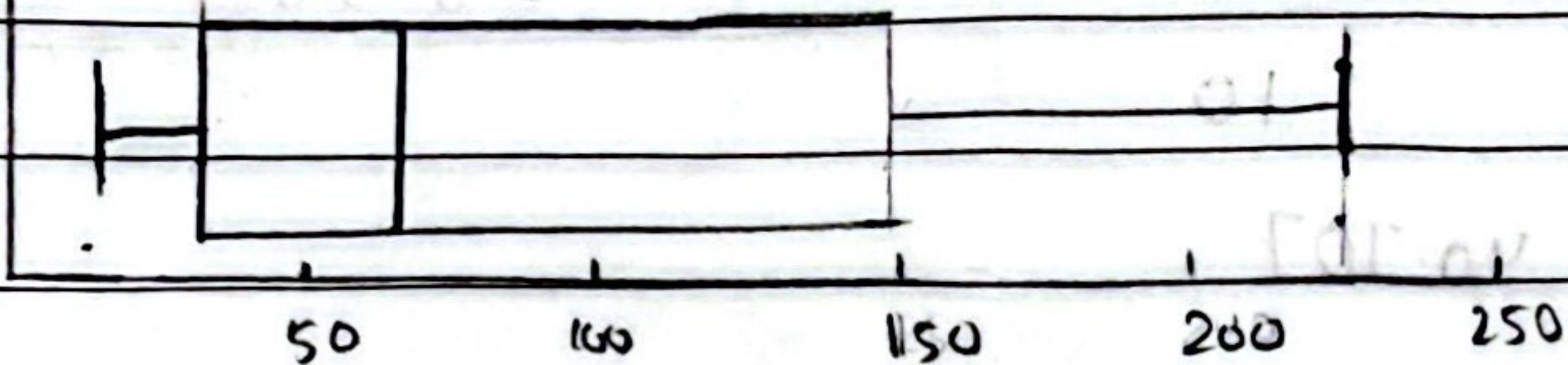
$$\text{upper fence} = 149.28 + 1.5 \times 117.36 = 325.32$$

Any value < -144.12 or > 325.32 is an outlier

No outlier detected.

Date _____

Boxplot of NO



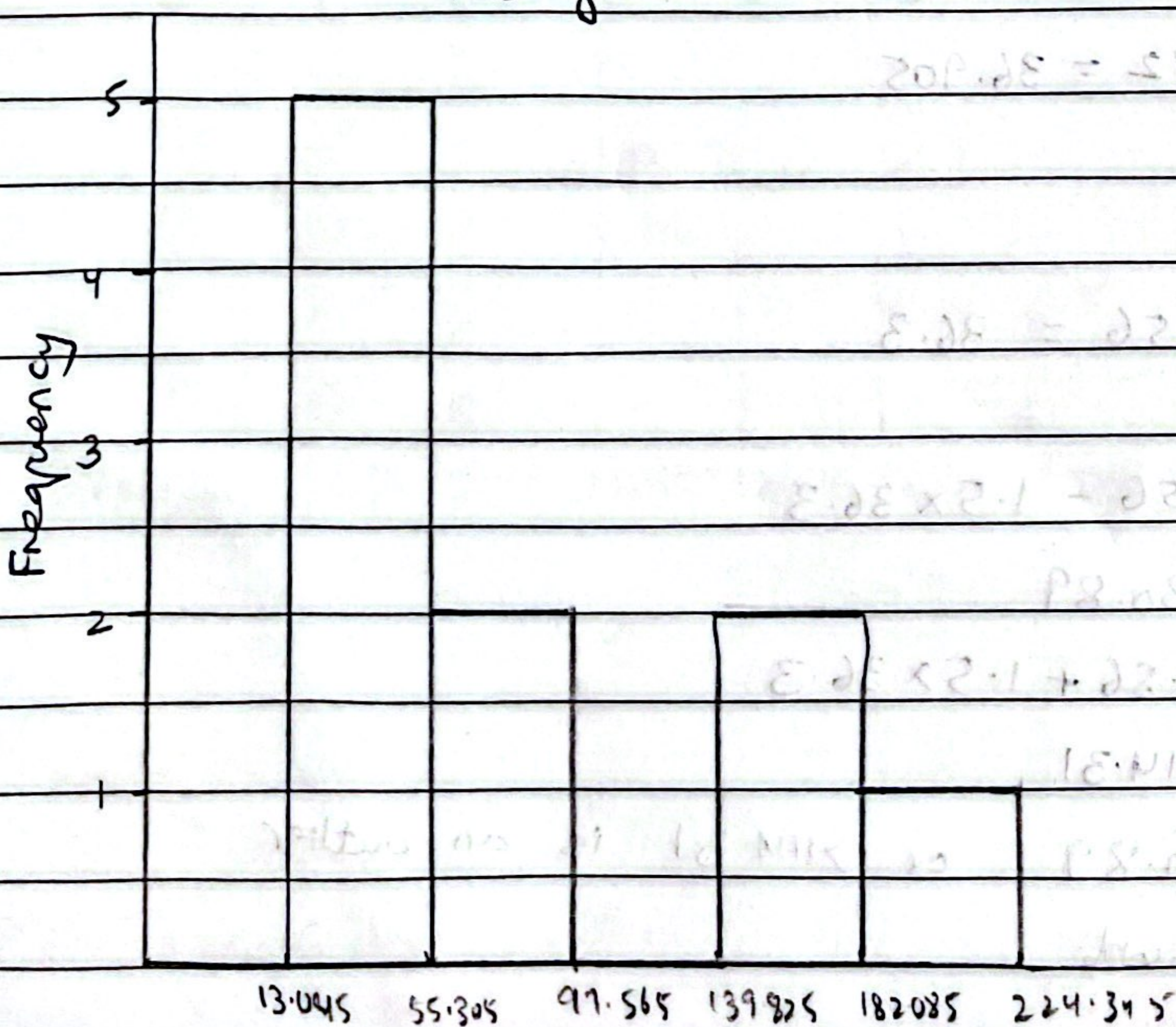
Class Limits	Class Boundaries	Frequency
13.05 - 55.30	13.045 - 55.305	5
55.31 - 97.56	55.305 - 97.565	2
97.57 - 139.82	97.565 - 139.825	0
139.83 - 182.08	139.825 - 182.085	2
182.09 - 224.34	182.085 - 224.345	1

$$C.W = 224.31 - 13.05$$

5

$$P = 42.252$$

Histogram of NO



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PM_{2.5} (Particulate Matter)

$$\text{mean} = \frac{32.32 + 77.31 + 59.86 + 37.21 + 60.49 + 48.42 + 16.71 + 14.59 + 23.56 + 36.60}{10}$$

$$\text{mean} = 40.707$$

$$\begin{aligned} \text{Std} = & (32.32 - 40.707)^2 + (77.31 - 40.707)^2 + (59.86 - 40.707)^2 + (37.21 - 40.707)^2 \\ & + (60.49 - 40.707)^2 + (48.42 - 40.707)^2 + (16.71 - 40.707)^2 + \\ & (14.59 - 40.707)^2 + (23.56 - 40.707)^2 + (36.60 - 40.707)^2 \end{aligned}$$

$$\text{std} = \sqrt{\frac{3808.864}{9}} = 20.572$$

Sorted: 14.59, 16.71, 23.56, 32.32, 36.60, 37.21, 48.42, 59.86, 60.49, 77.31

$$\text{min} = 14.59$$

$$Q_1 = 23.56$$

$$Q_2 = \frac{36.60 + 37.21}{2} = 36.905$$

$$Q_3 = 59.86$$

$$\text{max} = 77.31$$

$$\text{IQR} = 59.86 - 23.56 = 36.3$$

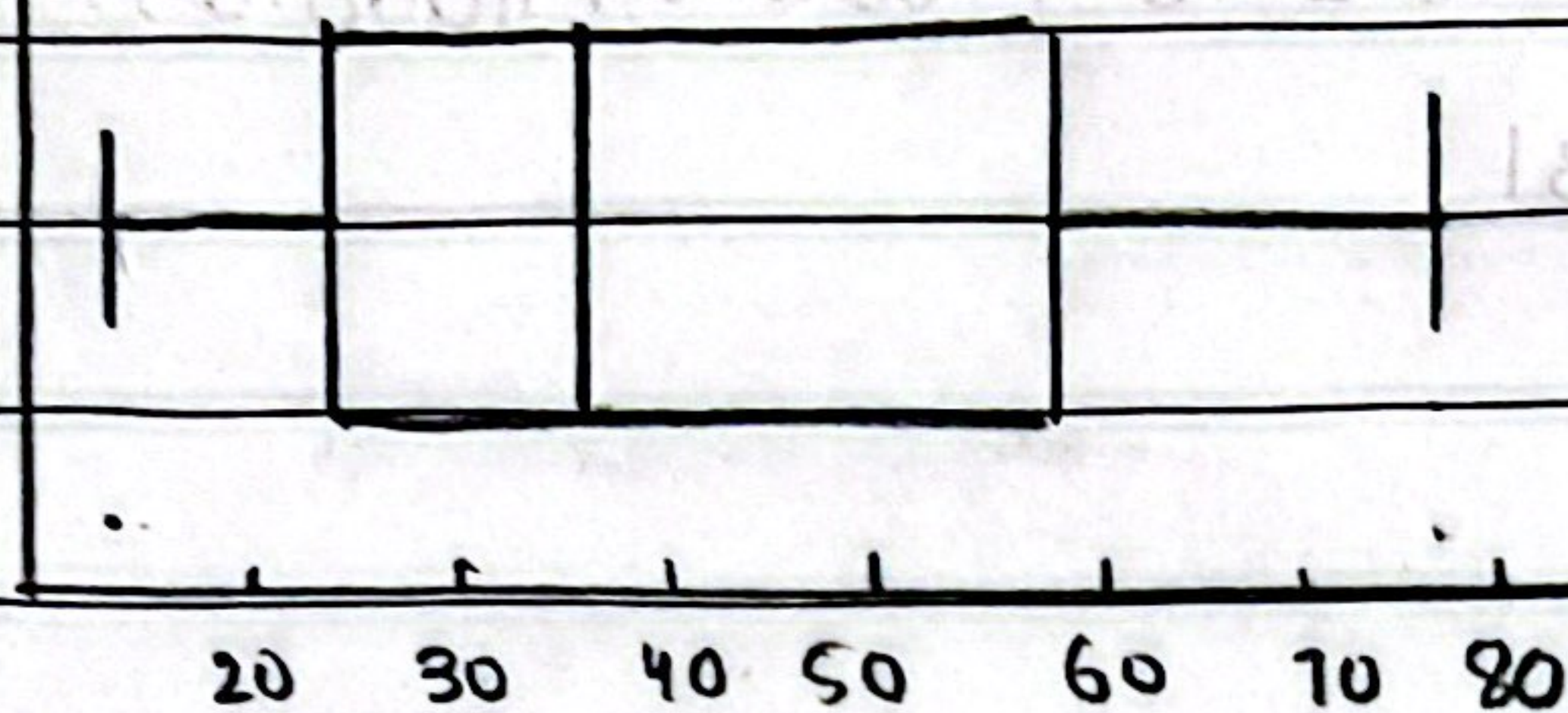
$$\begin{aligned} \text{Lower fence} &= 23.56 - 1.5 \times 36.3 \\ &= -30.89 \end{aligned}$$

$$\begin{aligned} \text{Upper fence} &= 59.86 + 1.5 \times 36.3 \\ &= 114.31 \end{aligned}$$

Any value < -30.89 or > 114.31 is an outlier

No outlier found

Boxplot of $PM_{2.5}$



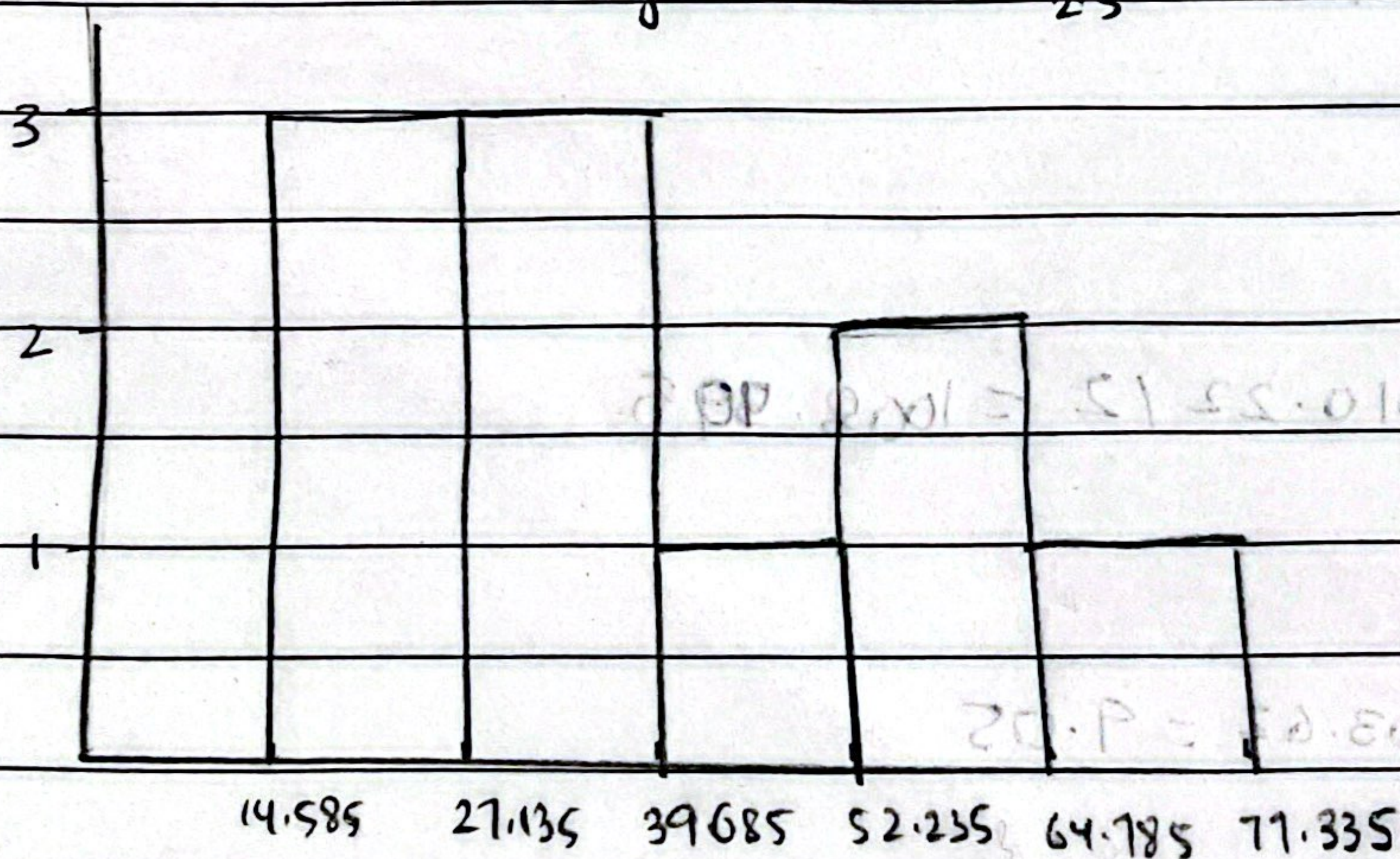
Class Limits	Class Boundaries	Frequency
14.59 - 27.13	14.585 - 27.135	3
27.14 - 39.68	27.135 - 39.685	3
39.69 - 52.23	39.685 - 52.235	1
52.24 - 64.78	52.235 - 64.785	2
64.79 - 77.33	64.785 - 77.335	1

$$C.W = 77.31 - 14.59$$

5

$$= 12.544 \approx 12.55$$

Histogram of $PM_{2.5}$



$$15 \cdot PPP = 0.10$$

$$52.8001 = 1.07$$

$$2 \cdot PPP = 51.55.0101 + 12.0001 = 52.75$$

$$12.5101 = 8.07$$

$$18.5101 = 8.00$$

$$20 \cdot P = 52.8001 - 12.5101 = 40.29$$

$$200057 \cdot 10.17.17$$

$$20 \cdot P \cdot 0.0.P + 52.8001 = 200057$$

$$20 \cdot 0.0.PP =$$

$$20 \cdot P \times 2.1 + 12.5101 = 2001$$

$$20 \cdot 5.2501 =$$

$$with no of 20 \cdot 5.2501 < 10 \quad 20 \cdot 0.0.PP > 10$$

$$10 \cdot 10.17.17$$

P (Pressure)

$$\text{mean} = \frac{1010.22 + 1012.67 + 1012.87 + 1012.73 + 1006.57 + 1000.63 + 994.21 + 1003.62 + 1010.94 + 1004.31}{10}$$

$$\text{mean} = 1006.877$$

$$\begin{aligned} \text{std} = & \frac{(1010.22 - 1006.877)^2 + (1012.67 - 1006.877)^2 + (1012.87 - 1006.877)^2}{9} \\ & + (1012.73 - 1006.877)^2 + (1006.57 - 1006.877)^2 + (1000.63 - 1006.877)^2 \\ & + (994.21 - 1006.877)^2 + (1003.62 - 1006.877)^2 + (1010.94 - 1006.877)^2 \\ & + (1004.31 - 1006.877)^2 \end{aligned}$$

$$\text{std} = \sqrt{\frac{348.184}{9}} = 6.220$$

Sorted: 994.21, 1000.63, 1003.62, 1004.31, 1006.57, 1010.22, 1010.94, 1012.67, 1012.73, 1012.87

$$\text{min} = 994.21$$

$$Q1 = 1003.62$$

$$Q2 = \frac{1006.57 + 1010.22}{2} = 1008.395$$

$$Q3 = 1012.67$$

$$\text{max} = 1012.87$$

$$\text{IQR} = 1012.67 - 1003.62 = 9.05$$

Outlier Fences

$$\begin{aligned} \text{Lower fence} &= 1003.62 - 1.5 \times 9.05 \\ &= 990.045 \end{aligned}$$

$$\begin{aligned} \text{Upper fence} &= 1012.67 + 1.5 \times 9.05 \\ &= 1026.245 \end{aligned}$$

Any value < 990.045 or > 1026.245 is an outlier
No outlier found

Date _____

Boxplot of P



995.10 997.5 1000 1008.5 1000.5 1007.5 1010 1012.5

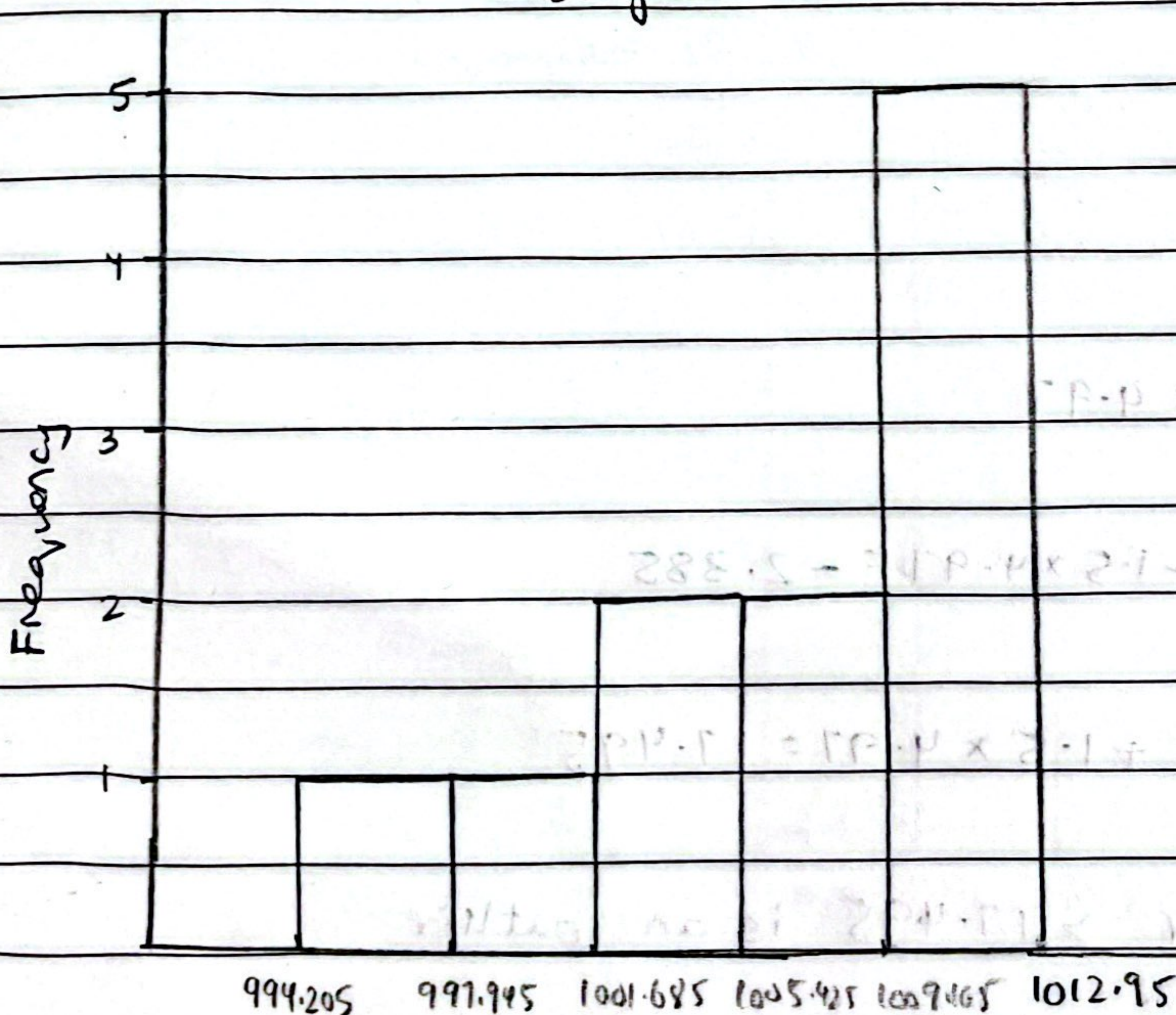
Class Limits	Class Boundaries	f.
994.21-997.94	994.205-997.945	1
997.95-1001.68	997.945-1001.685	1
1001.69-1005.42	1001.685-1005.425	2
1005.43-1009.16	1005.425-1009.165	1
1009.17-1012.9	1009.165-1012.95	5

$$C.W = 1012.87 - 994.21$$

5

$$C.W = 3.792$$

Histogram of P



Date _____

T (Temperature)

$$\text{mean} = \frac{6.20 + 8.79 + 10.04 + 11.14 + 9.09 + 9.81 + 13.61 + 5.07 + 3.03 + 1.90}{10}$$

$$\text{mean} = 7.868$$

$$\begin{aligned} \text{std} = & (6.20 - 7.868)^2 + (8.79 - 7.868)^2 + (10.04 - 7.868)^2 + (11.14 - 7.868)^2 \\ & + (9.09 - 7.868)^2 + (9.81 - 7.868)^2 + (13.61 - 7.868)^2 \\ & + (5.07 - 7.868)^2 + (3.03 - 7.868)^2 + (1.90 - 7.868)^2 \end{aligned}$$

$$\text{std} = \sqrt{\frac{124.144}{9}} = 3.714$$

Sorted: 1.90, 3.03, 5.07, 6.20, 8.79, 9.09, 9.81, 10.04, 11.14, 13.61

$$\text{min} = 1.90$$

$$Q1 = 5.07$$

$$Q2 = 8.94$$

$$Q3 = 10.04$$

$$\text{max} = 13.61$$

$$\text{IQR} = 10.04 - 5.07 = 4.97$$

Outlier Fences

$$\text{Lower fence} = 5.07 - 1.5 \times 4.97 = -2.385$$

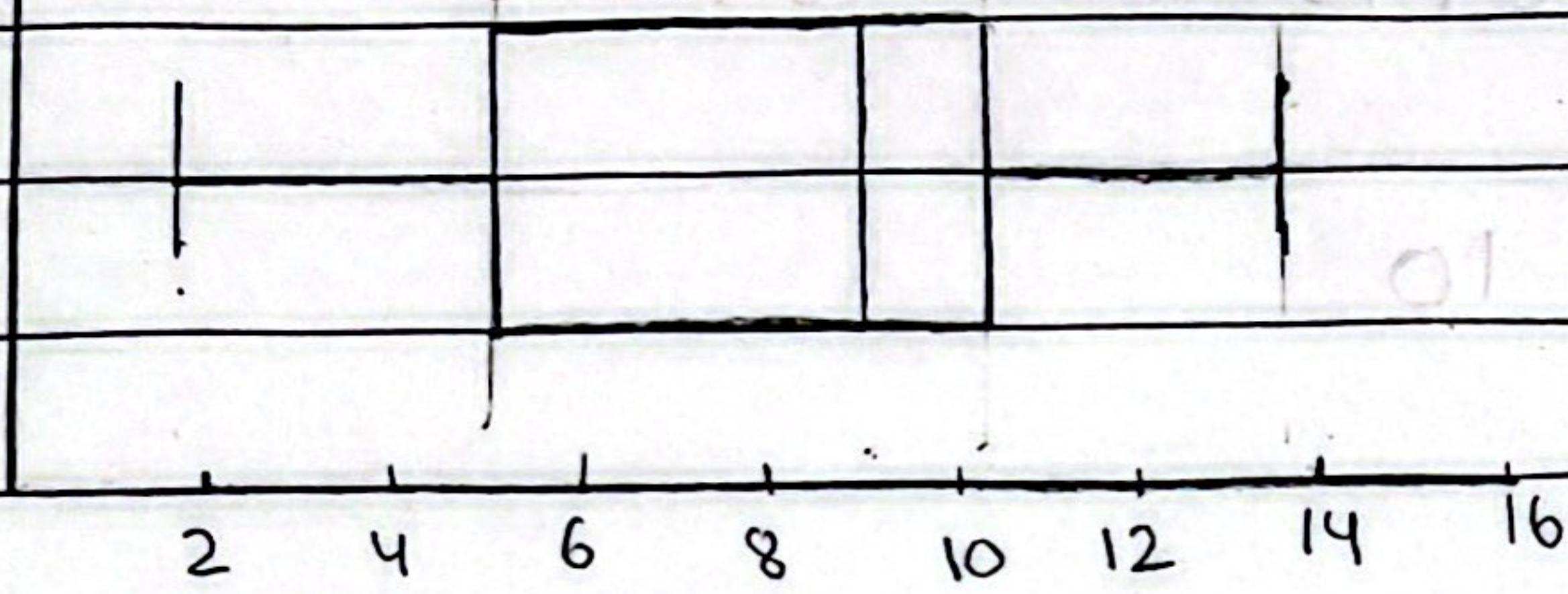
$$\text{Upper fence} = 10.04 + 1.5 \times 4.97 = 17.495$$

Any value < -2.385 or > 17.495 is an outlier

• No outlier found.

(Relative frequency)

Boxplot of T

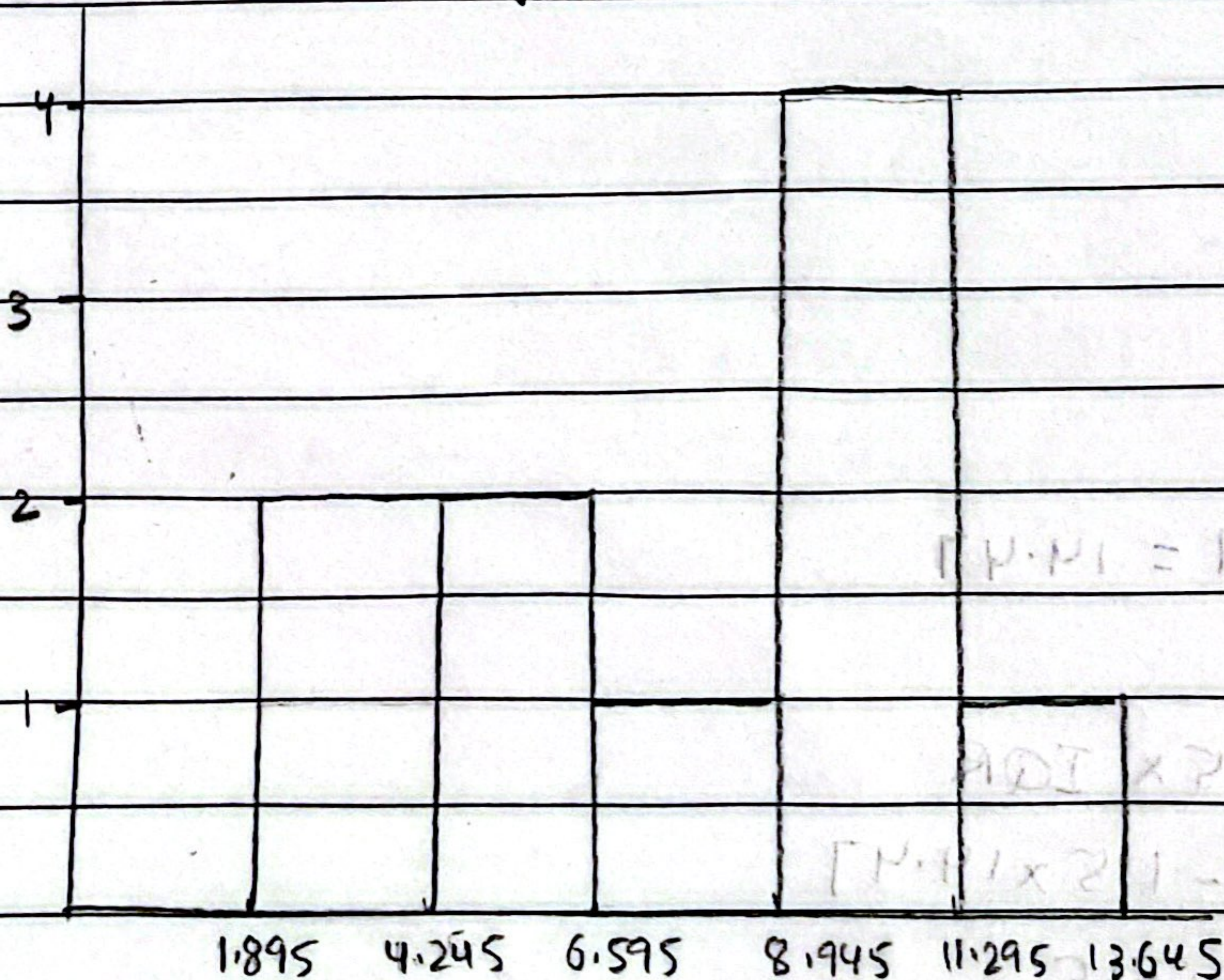


Class Limits	Class Boundaries	f
1.90 - 4.24	1.895 - 4.245	2
4.25 - 6.59	4.245 - 6.595	2
6.60 - 8.94	6.595 - 8.945	1
8.95 - 11.29	8.945 - 11.295	4
11.30 - 13.64	11.295 - 13.645	1

$$C.W = \frac{13.61 - 1.90}{5}$$

$$= 11.71 / 5 = 2.342$$

Histogram of T



10.00, 20.00

12.00 - 14.00

10.00 - 12.00

20.00 - 22.00

10.00 - 12.00

10.00 - 12.00

10.00 - 12.00

10.00 - 12.00

10.00 - 12.00

10.00 - 12.00

10.00 - 12.00

1.895 4.245 6.595 8.945 11.295 13.645

10.00 - 12.00

10.00 - 12.00

Date _____

RH (Relative Humidity)

$$\text{mean} = \frac{90.07 + 82.54 + 82.36 + 75.14 + 77.98 + 68.07 + 57.51 + 80.41 + 61.34 + 89.08}{10}$$

$$\text{mean} = 76.450$$

$$\text{Std} = \frac{(90.07 - 76.450)^2 + (82.54 - 76.450)^2 + (82.36 - 76.450)^2 + (75.14 - 76.450)^2 + (77.98 - 76.450)^2 + (68.07 - 76.450)^2 + (57.51 - 76.450)^2 + (80.41 - 76.450)^2 + (61.34 - 76.450)^2 + (89.08 - 76.450)^2}{10 - 1}$$

$$= \frac{1094.035}{9} = 11.0254$$

Sorted: 57.51, 61.34, 68.07, 75.14, 77.98, 80.41, 82.36, 82.54, 89.08, 90.07

$$\text{min} = 57.51$$

$$Q1 = 68.07$$

$$Q2 = 79.195$$

$$Q3 = 82.54$$

$$\text{max} = 90.07$$

$$\text{IQR} = 82.54 - 68.07 = 14.47$$

Outlier Fences

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR}$$

$$68.07 - 1.5 \times 14.47$$

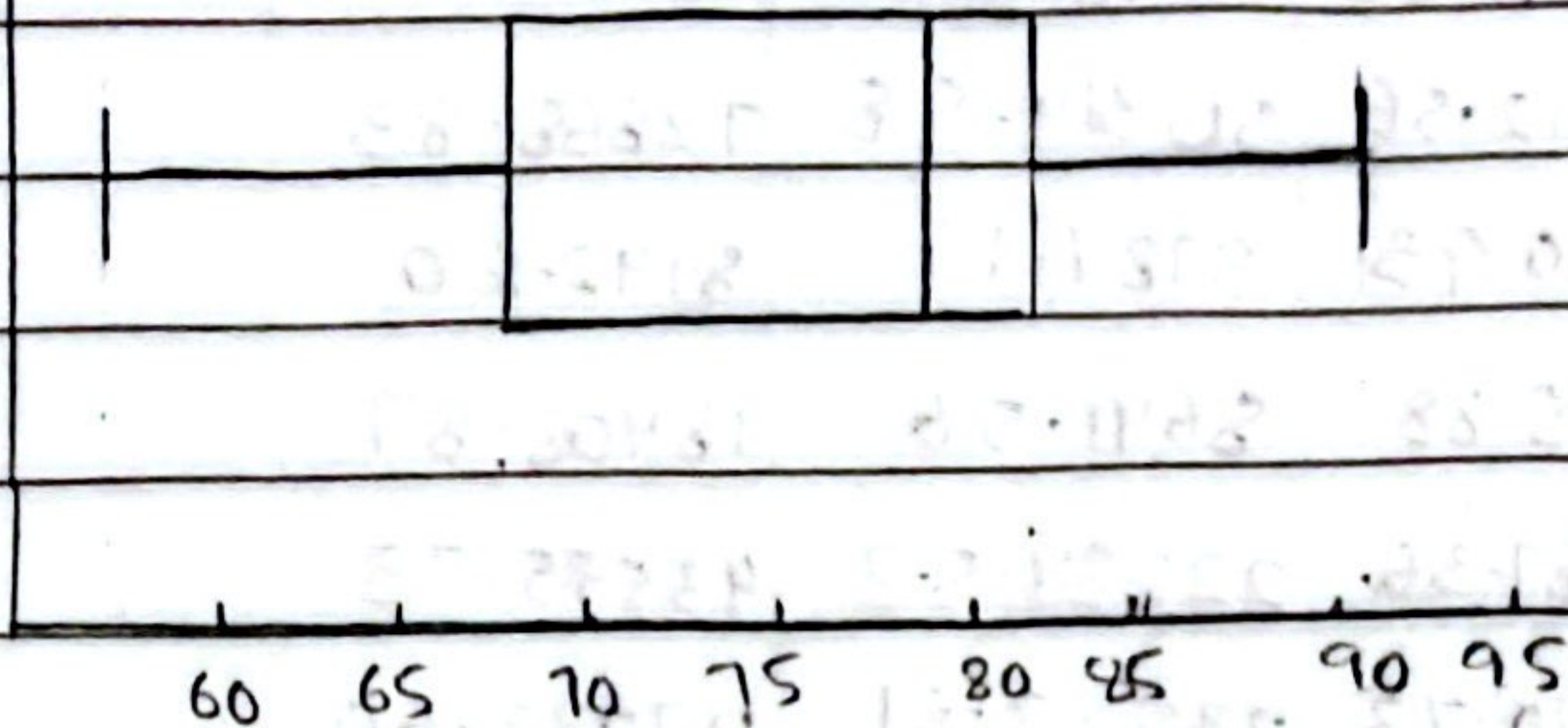
$$= 46.365$$

$$\text{upper fence} = Q3 + 1.5 \times \text{IQR} = 82.54 + 1.5 \times 14.47 = 104.245$$

No outlier found

Date _____

Boxplot of RH



Class Limits	Class Boundaries	f
57.51-64.02	57.505-64.025	2
64.03-70.54	64.025-70.545	1
70.55-77.06	70.545-77.065	1
77.07-83.58	77.065-83.585	4
83.59-90.1	83.585-90.105	2

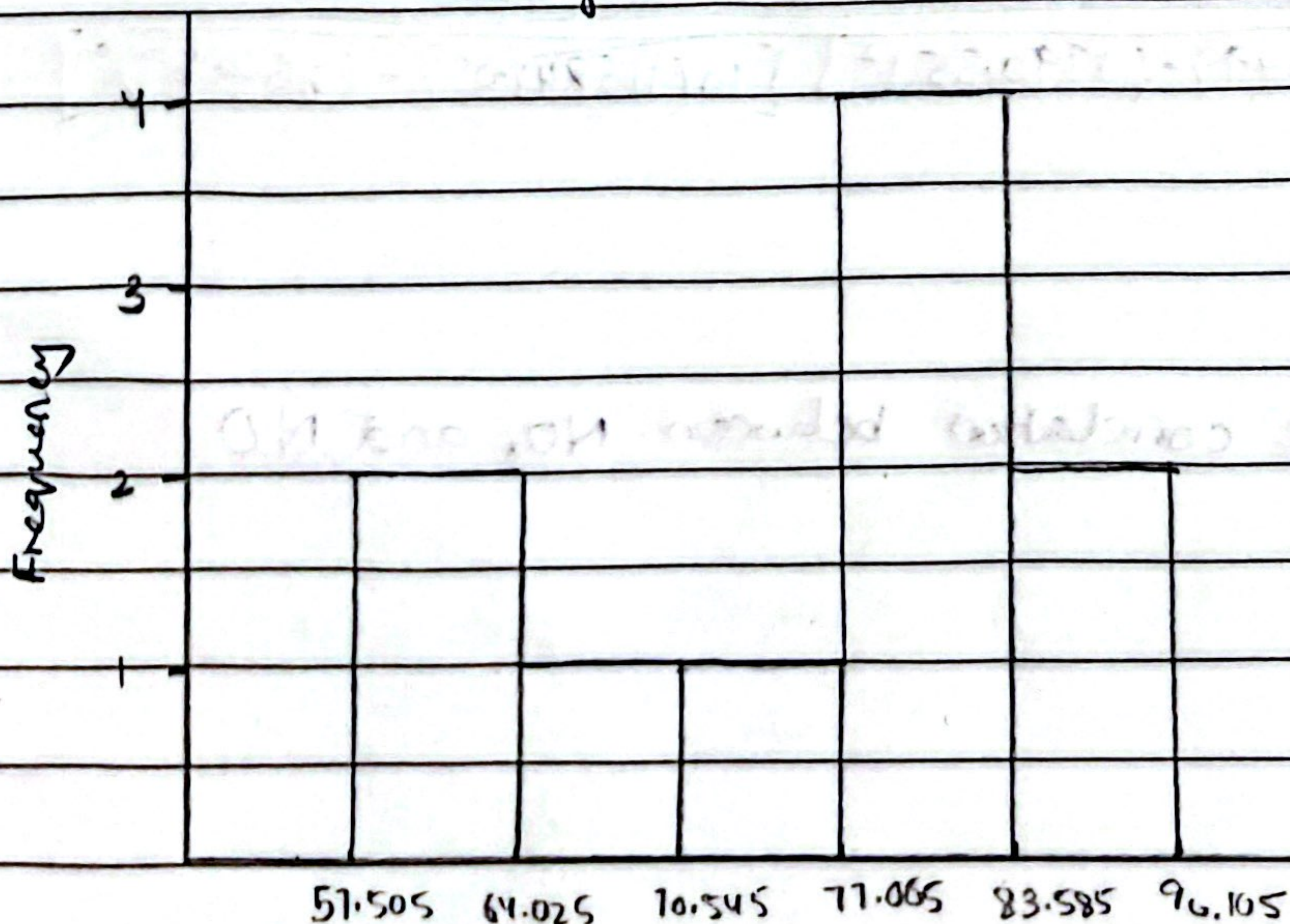
$$C.W = 90.07 - 57.51$$

$$= 32.56$$

$$\div 5$$

$$= 6.512$$

Histogram of RH



Date _____

NO_x and NO

14th July 2008

$\text{NO}_x (x)$	$\text{NO} (y)$	x^2	y^2	xy
127.47	53.37	16248.60	2848.36	6803.07
414.02	224.31	171412.56	50314.98	92868.83
133.12	61.54	17720.93	3787.17	8192.20
198.03	92.96	39215.88	8641.56	18408.87
292.04	149.28	85281.36	22284.52	43595.73
304.09	154.67	92470.73	23922.81	47033.60
123.45	47.76	15239.90	2281.02	5895.97
69.96	24.03	4894.40	577.44	1681.14
46.63	13.05	2174.36	170.30	608.52
81.54	31.92	6648.77	1018.89	2602.76

$$\sum x = 1790.35 \quad \sum y = 852.89 \quad \sum x^2 = 451313.49 \quad \sum y^2 = 115847.05 \quad \sum xy = 227690.69$$

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$$\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}$$

$$= \frac{10(227690.69) - (1790.35)(852.89)}{\sqrt{[10(451313.49) - (1790.35)^2][10(115847.05) - (852.89)^2]}}$$

$$\sqrt{[10(451313.49) - (1790.35)^2][10(115847.05) - (852.89)^2]}$$

$$r = 0.999$$

strong positive correlation between NO_x and NO